

Module 8. Deep Learning

- ❑ Motivation 
- ❑ Improve Training of Deep Learning Models
- ❑ Convolutional Neural Networks
- ❑ Recurrent Neural Networks
- ❑ Summary

Success of Deep Learning

The New York Times

Scientists See Promise in Deep-Learning Programs



Hao Zhang/The New York Times

A voice recognition program translated a speech given by Richard F. Rashid, Microsoft's top scientist, into Mandarin Chinese.

By JOHN MARKOFF

Published: November 23, 2012

Using an artificial intelligence technique inspired by theories about how the brain recognizes patterns, technology companies are reporting startling gains in fields as diverse as computer vision, speech recognition and the identification of promising new molecules for designing drugs.

 FACEBOOK

 TWITTER

 GOOGLE+

 SAVE



Success of Deep Learning



10 BREAKTHROUGH TECHNOLOGIES 2013

Deep Learning

With massive amounts of computational power, machines can now recognize objects and translate speech in real time. Artificial intelligence is finally getting smart. →

Temporary Social Media

Messages that quickly self-destruct could enhance the privacy of online communications and make people freer to be spontaneous. →

Prenatal DNA Sequencing

Reading the DNA of fetuses will be the next frontier of the genomic revolution. But do you really want to know about the genetic problems or musical aptitude of your unborn child? →

Memory Implants

A maverick neuroscientist believes he has deciphered the code by which the brain

Smart Watches

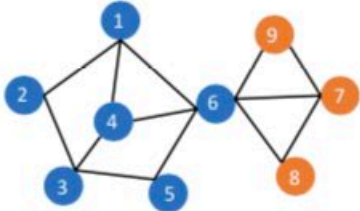
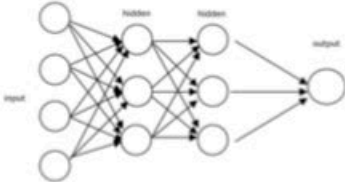
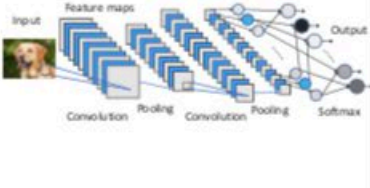
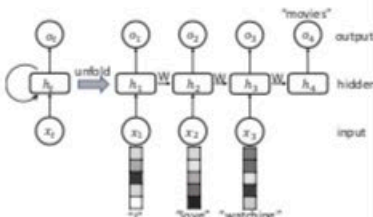
Ultra-Efficient Solar Power

Doubling the efficiency of a solar cell would completely

From Neural Networks to Deep Learning

- ❑ **Deep Learning** refers to training (deep) neural networks with many layers
 - ❑ More neurons, more layers
 - ❑ More complex ways to connect layers
- ❑ Deep Learning has some major advantages, making it popular
 - ❑ Tremendous improvement of performance in many tasks
 - ❑ Image recognition, natural language processing, AI game playing...
 - ❑ Requires no (or less) feature engineering, making end-to-end models possible
- ❑ Several factors lead to deep learning's success
 - ❑ Very large data sets
 - ❑ Massive amounts of computation power (GPU acceleration)
 - ❑ Advanced neural network structures and tricks
 - ❑ Convolutional neural networks, recurrent neural networks, graph convolution network ...
 - ❑ Dropout, ReLU, residual connection, ...

Overview of Typical Deep Learning Architectures

Data type	Multi-dimensional Features: credit rating, account balance $x = (4.5, 500, 3, 5)$ #deposits, #withdraws	Grid 	Sequence $x = \text{"I love watching movies."}$	Graph 
DL Architecture	Feed-forward Network 	CNN 	RNN 	GNN 